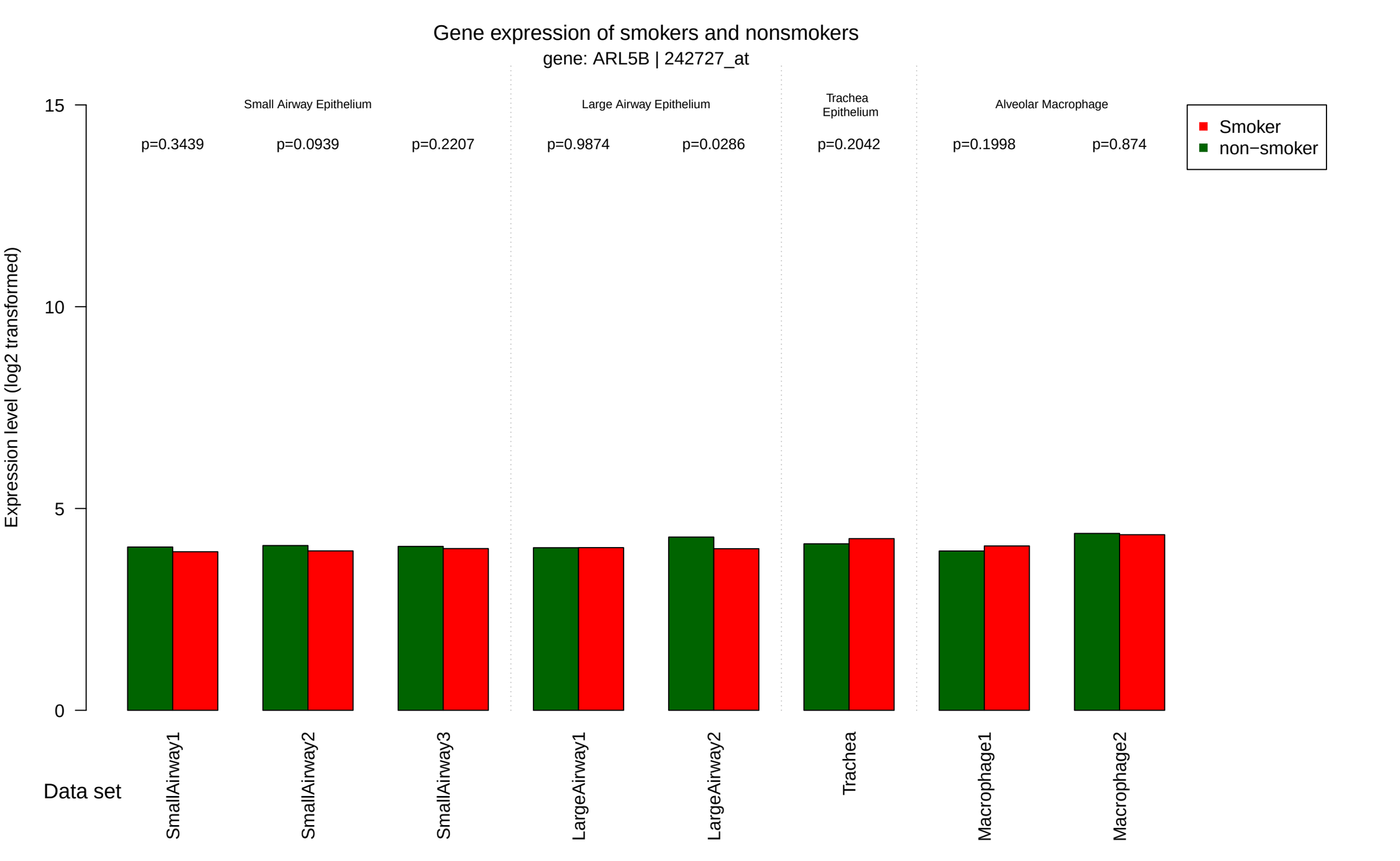
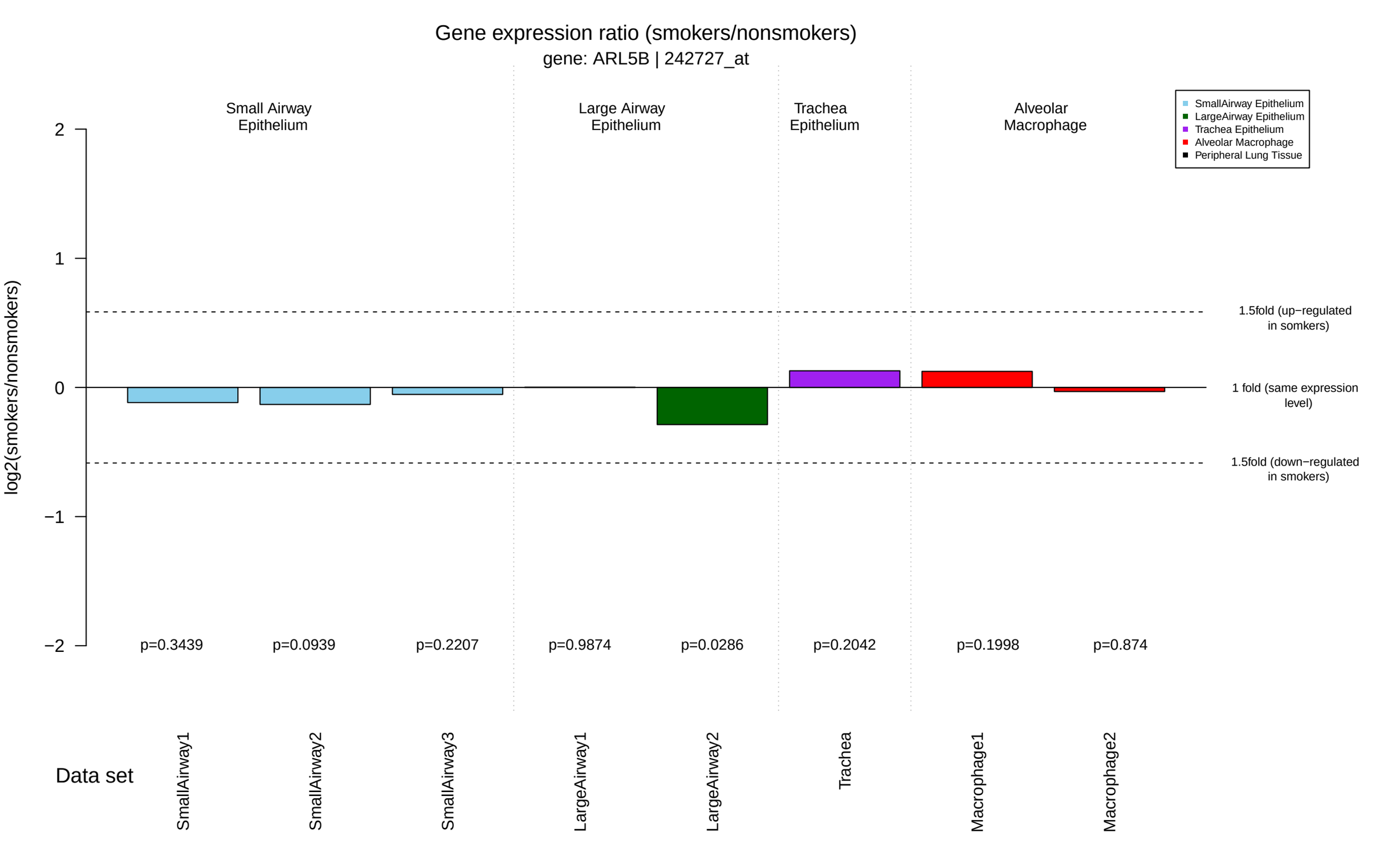
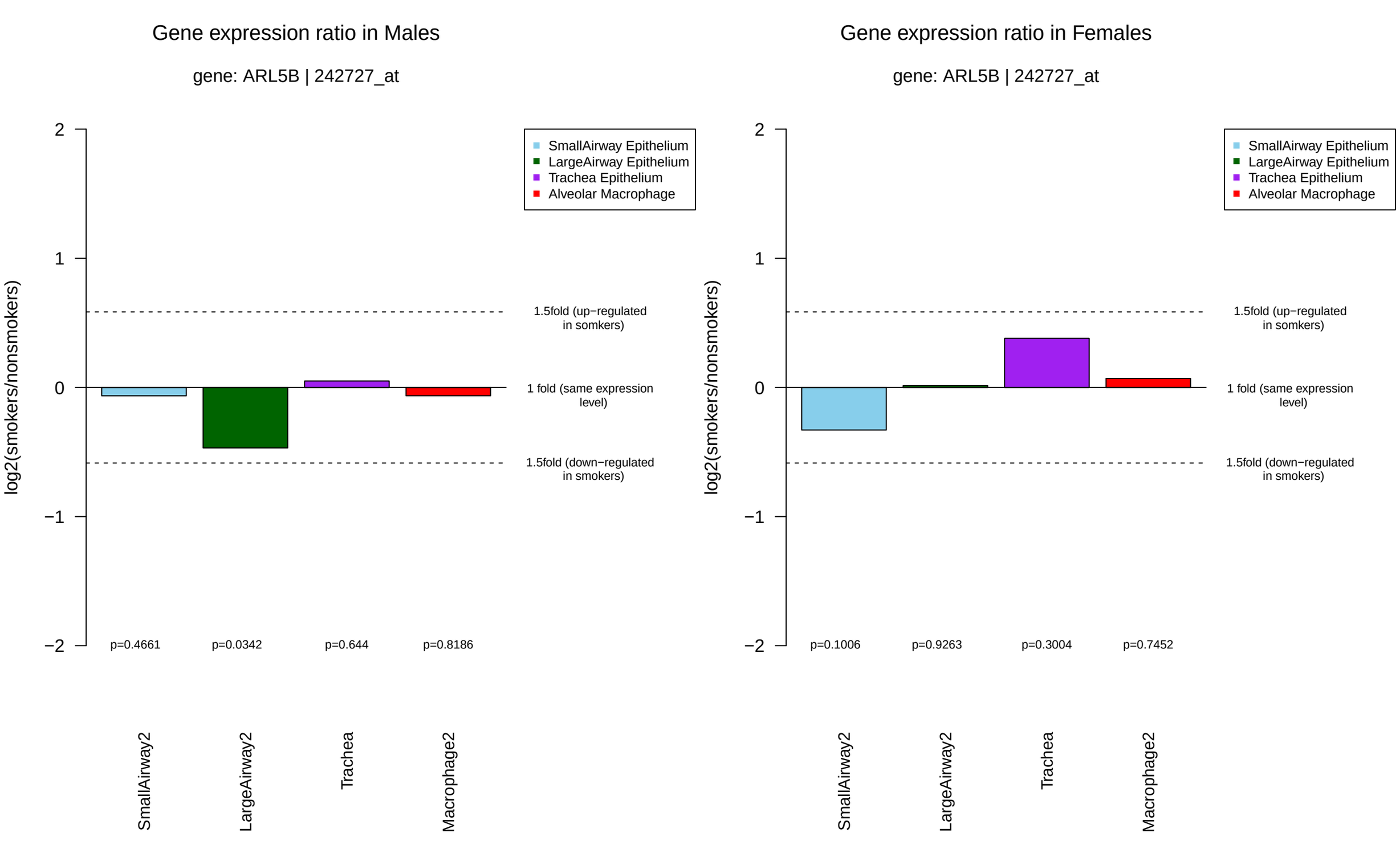
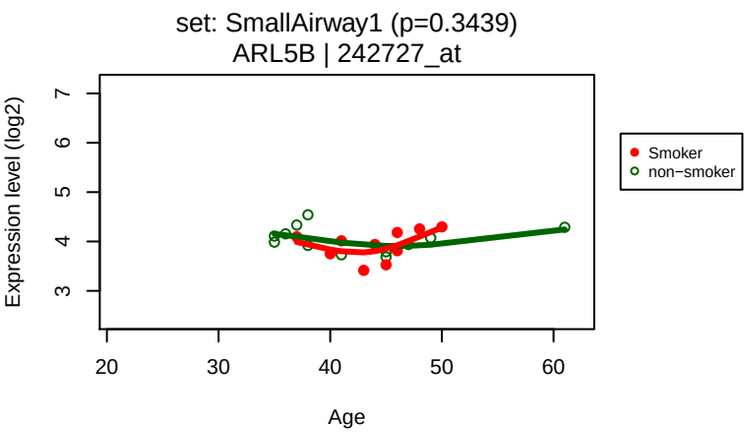
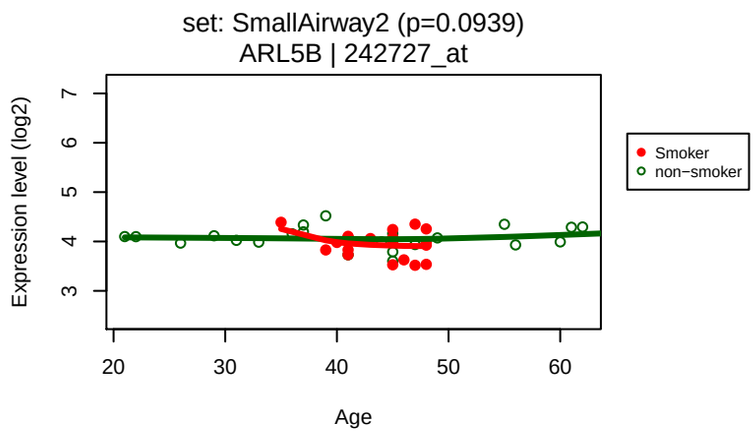
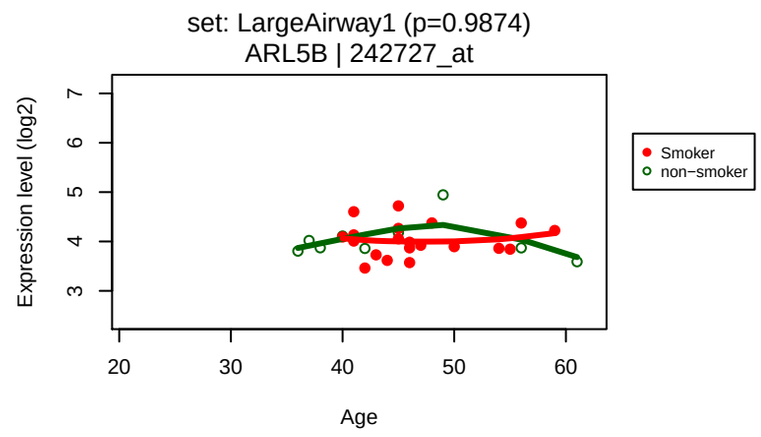

Gene expression of smokers and nonsmokers
gene: ARL5B | 242727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3439
p=0.0939
p=0.2207
p=0.9874
p=0.0286
p=0.2042
p=0.1998
p=0.874
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

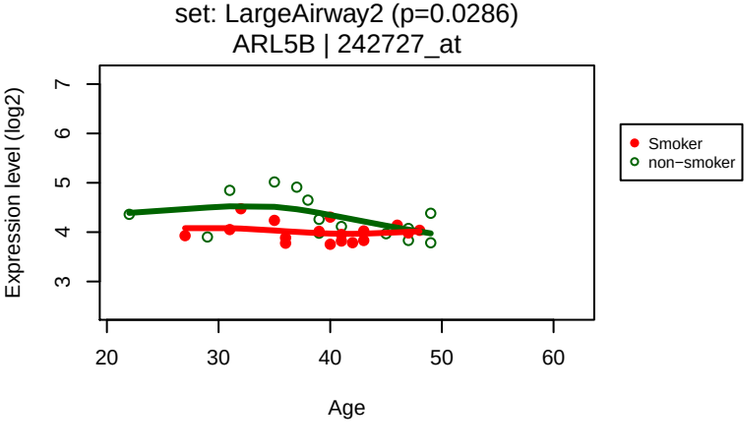

Gene expression ratio (smokers/nonsmokers)
gene: ARL5B | 242727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3439
p=0.0939
p=0.2207
p=0.9874
p=0.0286
p=0.2042
p=0.1998
p=0.874
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

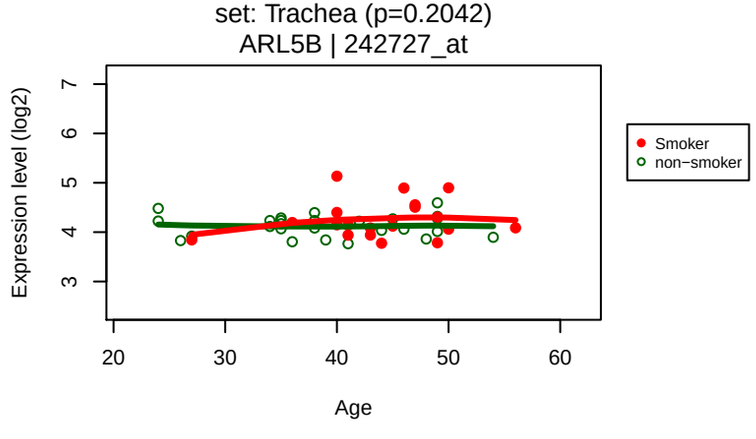

Gene expression ratio in Males
gene: ARL5B | 242727_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4661
p=0.0342
p=0.644
p=0.8186
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL5B | 242727_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1006
p=0.9263
p=0.3004
p=0.7452
SmallAirway2
LargeAirway2
Trachea
Macrophage2

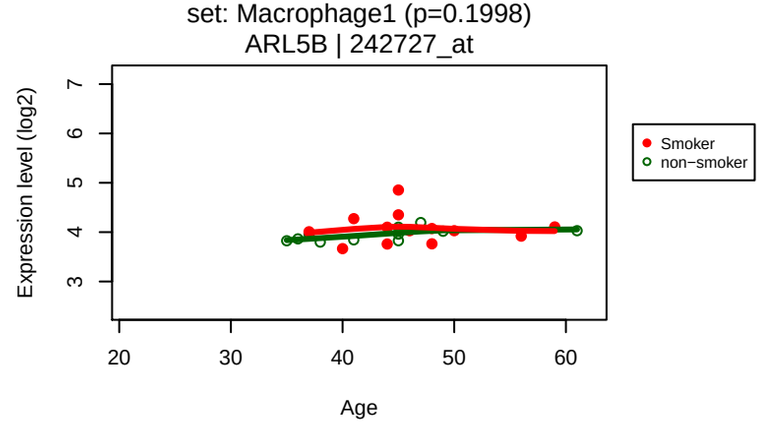

set: SmallAirway1 (p=0.3439)
ARL5B | 242727_at
Expression level (log2)
Age
Smoker
non−smoker

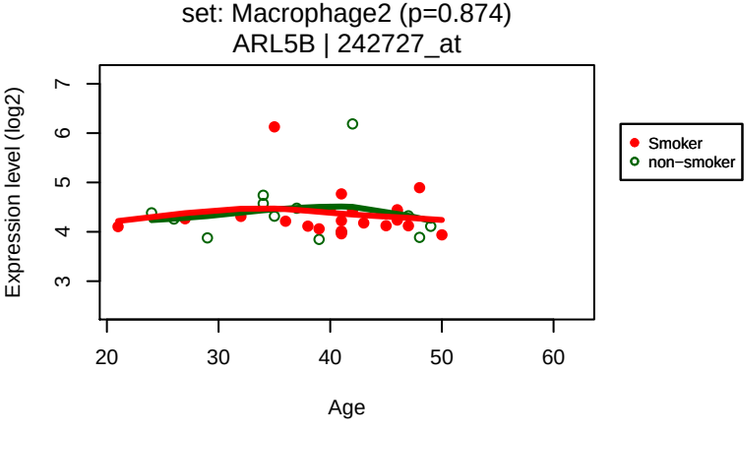

set: SmallAirway2 (p=0.0939)
ARL5B | 242727_at
Expression level (log2)
Age
Smoker
non−smoker

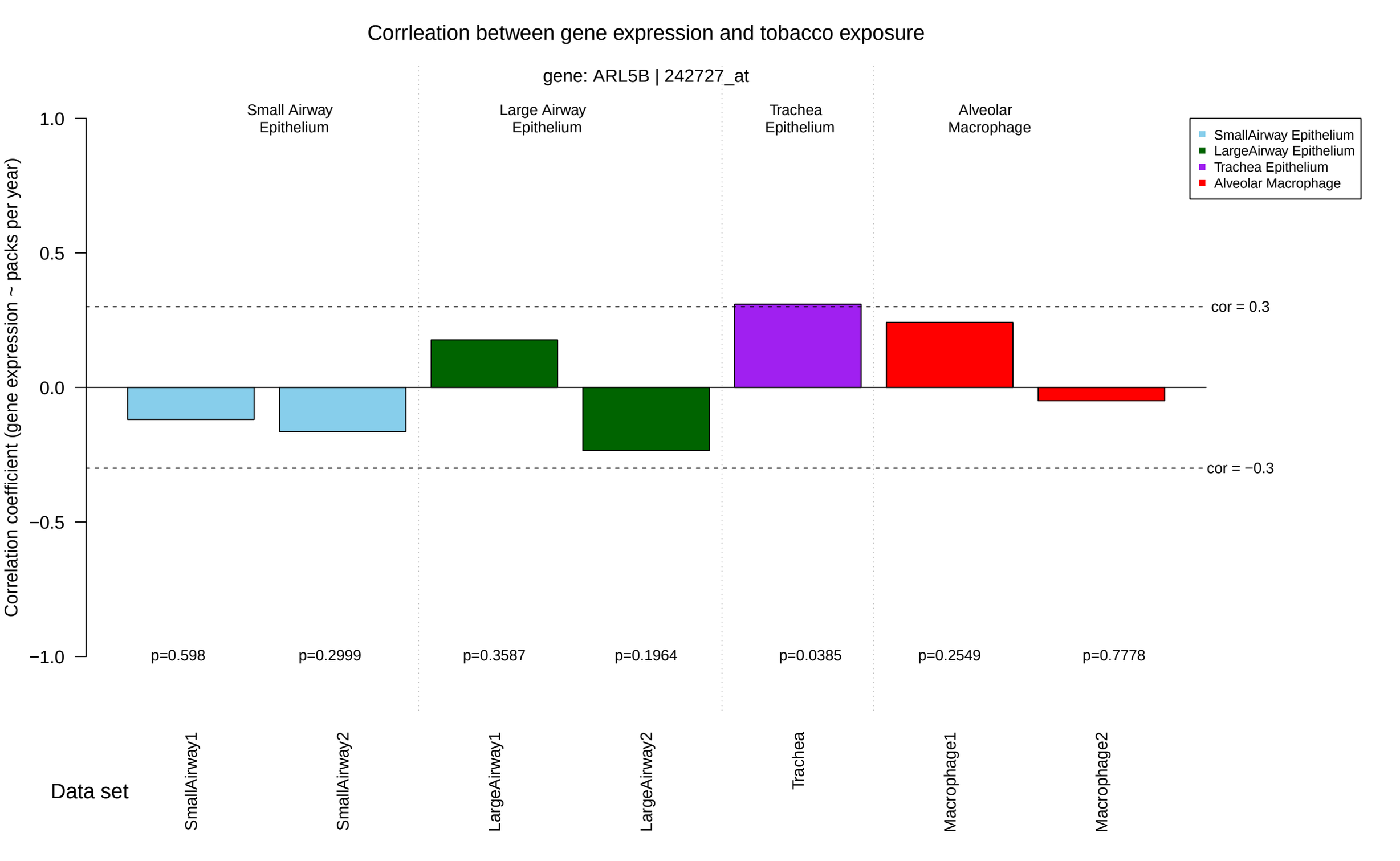

set: LargeAirway1 (p=0.9874)
ARL5B | 242727_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0286)
ARL5B | 242727_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2042)
ARL5B | 242727_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1998)
ARL5B | 242727_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.874)
ARL5B | 242727_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL5B | 242727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.598
p=0.2999
p=0.3587
p=0.1964
p=0.0385
p=0.2549
p=0.7778
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2